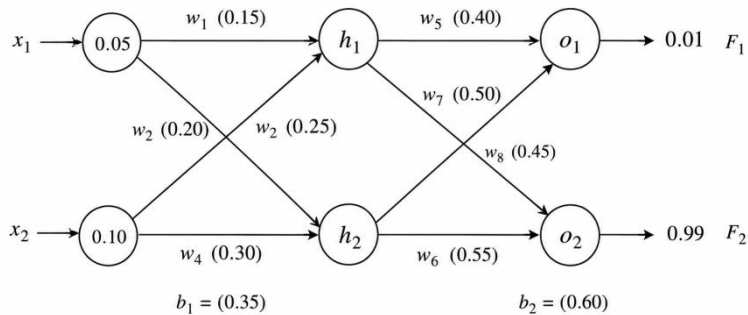


20CS41008–Deep Learning
(Assignment 1 Questions)

UNIT	Questions	CO	BTL
1	1. Using Threshold Logic, design McCulloch-Pitts (MCP) neurons to implement the AND, OR, and NOT logical gates. Specify the input and threshold for each neuron and verify the output for all possible input combinations.	1	3
	2) Given a perceptron has initial weights $w_1 = 0.2$, $w_2 = -0.3$, bias 0.1, learning rate 0.2. For input (1,1) with target output 1, perform one iteration of the Perceptron Learning Algorithm and calculate the updated weights and bias.	1	3
2	1) Compute one complete iteration of the Backpropagation algorithm using the feed-forward neural network given below.. 	1	3
	2) Perform Eigenvalue Decomposition of $A = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$. And find eigen values along with their eigen vectors.	1	3
3	1. Given an Autoencoder with input $x = [1, 2, 3]^T$, encoder $h = g(Wx + b)$ and decoder, $\hat{x} = f(W'h + c)$, where the weights, biases, and activation functions are specified, calculate the hidden representation and reconstructed output. Also determine the reconstruction error. $W = \begin{bmatrix} 0.5 & 0.2 & -0.1 \\ 0.1 & -0.3 & 0.4 \end{bmatrix}$ $b = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$ $W' = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$ $c = \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix}$	2	4